CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                          struct nighttime {
        {                                       int  x, y,z;
        int hr,min,sec;                              }
        }
                struct {
              int hr,min,sec:
                    } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 – PROGRAMMING PARADIGMS
ASSESSMENT-II

TIME: 1 ½ hour CLASS : **VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART – A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for exceptions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

14/14

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 2 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

8

14

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
 while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour **CLASS :VI SEM – (P)** **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                         struct nighttime {
        {                                      int  x, y,z;
        int hr,min,sec;                              }
        }
              struct {
           int hr,min,sec:
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struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

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```
int x = 120;
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}
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void function3 () {
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function2();
}
function1();
function2();
function3();
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b. Classify the programming languages in detail. **(5)**

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b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

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b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART – A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

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7. a) Differentiate copy and reference semantics. **(5)**
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```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. 2 4

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8

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

14

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                         struct nighttime {
     {                                        int  x, y,z;
     int hr,min,sec;                               }
     }
            struct {
         int hr,min,sec:
               } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

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```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
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void function1 () {
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print(x + y);
}
void function2 () {
int x = 0;
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int y = 0;
function2();
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function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
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```
int h, i ;
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int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

14/14

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 2 4

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8

14

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS : VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                          struct nighttime {
      {                                        int  x, y,z;
      int hr,min,sec;                               }
      }
            struct {
         int hr,min,sec:
               } a, b;
struct daytime c, d;
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```
// compute the factorial of integer n
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int n, i, result;
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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE: 24/3/26** **MARKS: 50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
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PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. (5)
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void B(int w){
int j, k;
i = 2*w;
w = w+i;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6 2

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 4

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8

14

```
// compute the factorial of n
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   f=1;
 while (i < n){
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}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

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1. How will you check two types are equal?

```
struct daytime
    {
    int hr,min,sec;
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                                struct nighttime {
                                    int x, y,z;
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        struct {
    int hr,min,sec:
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struct daytime c, d;
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PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
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void function1 () {
```

```
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}
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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS: VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

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void A(int x, int y) {
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B( h);
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int a, b;
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Part-C

14/14

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     f = f * i ;
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CS23017 – PROGRAMMING PARADIGMS

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

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bool i,j ;
B( h);
}
int main()
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int a, b;
h=5;a= 3;b=2;
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```

14/14

Part-C

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void main ( ) {
    int  n, i, f;
    n = 3;
    i=1;
    f=1;
while (i < n){
     i= i + 1;
     f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour CLASS :VI SEM – (P) **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
    {
    int hr,min,sec;
    }
                                struct nighttime {
                                    int  x, y,z;
                                    }
        struct {
    int hr,min,sec:
            } a, b;

struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

5

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS
ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for exceptions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

16. }

2

14/14

Part-C

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

8

14

```
// compute the factorial of n
void main ( } {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS : VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                          struct nighttime {
    {                                        int  x, y,z;
    int hr,min,sec;                              }
    }
          struct {
        int hr,min,sec:
              } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
 }
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

5 b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 – PROGRAMMING PARADIGMS
ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/3/26** **MARKS:50**

Answer any 5 PART – A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for excempions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+i;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. 2 4
(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

8

```
// compute the factorial of n
void main ( ) {
   int n, i, f;
   n = 3;
   i=1;
   f=1;
 while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

14

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
        {
        int hr,min,sec;
        }
                                struct nighttime {
                                        int  x, y,z;
                                        }
            struct {
        int hr,min,sec:
              } a, b;

struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

5

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for exceptions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

14/14

6

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** (2+4)

2

4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

8

14

```
// compute the factorial of n
void main ( ) {
   int n, i, f;
   n = 3;
   i=1;
   f=1;
 while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS : VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                          struct nighttime {
    {                                        int  x, y,z;
    int hr,min,sec;                              }
    }
            struct {
         int hr,min,sec:
                } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
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2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

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PART – C (14 marks)

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/3/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
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9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
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```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

Part-C

14/14

6

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. (2+4)

2 4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: **(4+4)**

8

14

```
// compute the factorial of n
void main ( ) {
   int n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour CLASS :VI SEM – (P) **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
    {
    int hr,min,sec;
    }
                                struct nighttime {
                                        int x, y,z;
                                        }
        struct {
    int hr,min,sec:
        } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

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2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

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CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

2023103062

Answer any 5 PART – A (5 X 2 = 10)

1. What is semantics?
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PART – B (2 X 13 = 26 marks)

Answer any 2

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int h, i ;
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int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

2

Part-C

14/14

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** (2+4)

4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

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14

```
// compute the factorial of n
void main ( ) {
    int n, i, f;
    n = 3;
    i=1;
    f=1;
  while (i < n){
     i= i + 1;
     f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                    struct nighttime {
    {                                     int  x, y,z;
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```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
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PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
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void function1 () {
```

```
print(x + y);
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void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
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```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

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PART – C (14 marks)

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
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5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

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   b) Explain the classification of exceptions in java. **(5)**
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int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. (2+4)

2 4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

8

14

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour **CLASS :VI SEM – (P)** **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                     struct nighttime {
    {                                   int x, y,z;
    int hr,min,sec;                         }
    }
          struct {
        int hr,min,sec:
              } a, b;
struct daytime c, d;
struct nighttime e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour CLASS : **VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for excempions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all fom forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. 2 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

14 8

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour | **CLASS :VI SEM – (P)** | **DATE:4/2/26** | **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
    {
    int hr,min,sec;
    }

struct nighttime {
    int  x, y,z;
    }

        struct {
    int hr,min,sec:
            } a, b;

struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE:24/2/26** **MARKS:50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
   c) Explain the approaches used in defining the semantics? **(3)**
8. a) What is the support for streams for sequential I/O in java. **(5)**
   b) Define an exception. Give five examples for excempions at different levels of abstraction **(3+5)**
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all fom forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

14/14

Part-C

6

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B**. 2 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

14 8

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour CLASS :VI SEM – (P) **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime
    {
    int hr,min,sec;
    }

struct nighttime {
    int x, y,z;
    }

        struct {
      int hr,min,sec:
            } a, b;

struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
c) c and e

2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
4. Differentiate narrowing and widening conversion. Give suitable example.
5. What forms the dope vector? Give suitable example.
6. Construct the type map for the given code.

```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

(i) Mixing compilation and interpretation

(ii) Library routines and linking.

b. Bring out the equivalence of array and pointers in C with suitable code. **(7)**

9) a) How the various languages deal with wasting minimum amount of storage. Justify. **(6)**

b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

10) a) Justify the need for different programming languages giving 4 reasons. **(8)**

b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS: VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. (5)
   b) Explain the classification of exceptions in java. (5)
   c) Explain the approaches used in defining the semantics? (3)
8. a) What is the support for streams for sequential I/O in java. (5)
   b) Define an exception. Give five examples for excempions at different levels of abstraction (3+5)
9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. (2+5)
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area. (6)

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

16. }

Part-C

14/14

6 2

10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: (4+4)

14 8

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
 }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour CLASS :VI SEM – (P) **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                         struct nighttime {
    {                                          int x, y,z;
    int hr,min,sec;                                }
    }
            struct {
        int hr,min,sec:
                } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

a) c and d
b) a and c
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2. What are self-hosting compilers? Give suitable example.
3. Differentiate type and type error.
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```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
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```

```
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void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

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b) Write the formula for calculating the address of array elements in row and column major ordering. Also illustrate the storage of array elements (2d) in row and column major ordering. **(7)**

PART – C (14 marks)

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5 b) Differentiate statically, dynamically and strongly typed languages. Give suitable example. **(6)**

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CS23017 - PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS: VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART - A (5 X 2 = 10) 2023103062

1. What is semantics?
2. What is meant by partial function in semantics?
3. Which parameter passing mechanism is called eager evaluation? Why?
4. Draw the structure of called function's activation record and write the significance of static and dynamic link.
5. Find the Cambridge prefix notation for the below expression tree.

6. When is the programming language Turing complete.

PART - B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. **(5)**
   b) Explain the classification of exceptions in java. **(5)**
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9. a) What is side effect in an expression? Give the rule corresponding to meaning of an expression based on its all form forms. **(2+5)**
   b) Modify the code given below to achieve call by reference mechanism and diagrammatically show the memory layout of all the functions and static area **(6)**

```
int h, i ;
void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

16. }

Part-C

14/14

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.** 2 4

(2+4)

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: **(4+4)**

8

14

```
// compute the factorial of n
void main ( ) {
   int  n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```

CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-I

TIME: 1 ½ hour CLASS :VI SEM – (P) **DATE:4/2/26** **MARKS:50**

PART –A (5 X 2 = 10) Answer any 5

1. How will you check two types are equal?

```
struct daytime                          struct nighttime {
      {                                        int  x, y,z;
      int hr,min,sec;                               }
      }
            struct {
          int hr,min,sec:
                 } a, b;
struct daytime c, d;
struct nighttime  e;
```

Check whether the following are equal considering the above code. If equal, then find the type of equivalence:

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```
// compute the factorial of integer n
void main ( ) {
int n, i, result;
n = 8;
i = 1;
result= 1;
while (i <=n) {
i=i+l;
j=j+1;
}
}
```

PART – B (2 X 13 = 26 Marks) Answer any 2

7) a) Consider the following C-like program:

```
int x = 120;
int y = 45;
void function1 () {
```

```
print(x + y);
}
void function2 () {
int x = 0;
function1();
}
void function3 () {
int y = 0;
function2();
}
function1();
function2();
function3();
```

What does this program print if the language uses static scoping and dynamic scoping? Draw symbol table and explain. **(8)**

b. Classify the programming languages in detail. **(5)**

8) a) Explain the following **(6)**

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PART – C (14 marks)

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CS23017 – PROGRAMMING PARADIGMS

ASSESSMENT-II

TIME: 1 ½ hour **CLASS : VI SEM(N,P&Q)** **DATE: 24/2/26** **MARKS: 50**

Answer any 5 PART – A (5 X 2 = 10) 2023103062

1. What is semantics?
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6. When is the programming language Turing complete.

PART – B (2 X 13 = 26 marks)

Answer any 2

7. a) Differentiate copy and reference semantics. (5)
   b) Explain the classification of exceptions in java. (5)
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void B(int w){
int j, k;
i = 2*w;
w = w+1;
}
void A(int x, int y) {
bool i,j ;
B( h);
}
int main()
{
int a, b;
h=5;a= 3;b=2;
A(a,b);
```

```
}
```

2

14/14

Part-C

6 10.a) What is short circuit evaluation? Apply it to **A&B** and **A or B.**
(2+4) 4

b) Define program state and the maps associated with the state. Find the run-time trace for the code given below: **(4+4)**

8

14

```
// compute the factorial of n
void main ( ) {
   int n, i, f;
   n = 3;
   i=1;
   f=1;
while (i < n){
    i= i + 1;
    f = f * i ;
  }
}
```